

1.
sol.

Given that

$$\text{let } A = \begin{bmatrix} 1 & 1 & 3 & 5 \\ 2 & 2 & 1 & 3 \\ 4 & 4 & 7 & 3 \\ 2 & 4 & 3 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - 4R_1, R_4 \rightarrow R_4 - 2R_1$$

$$\sim \begin{bmatrix} 1 & 1 & 3 & 5 \\ 0 & 0 & -5 & -7 \\ 0 & 0 & -5 & -17 \\ 0 & 2 & -3 & -9 \end{bmatrix} \quad R_2 \leftrightarrow R_4$$

$$\sim \begin{bmatrix} 1 & 1 & 3 & 5 \\ 0 & 2 & -3 & -9 \\ 0 & 0 & -5 & -17 \\ 0 & 0 & -5 & -7 \end{bmatrix} \quad R_4 \rightarrow R_4 - R_3$$

$$\sim \begin{bmatrix} 1 & 1 & 3 & 5 \\ 0 & 2 & -3 & -9 \\ 0 & 0 & -5 & -17 \\ 0 & 0 & 0 & 10 \end{bmatrix}$$

$\therefore$ Which is in Echelon form

No. of non zero rows = 4

$$\therefore \boxed{\rho(A) = 4}$$

2.
sol.

Cauchy - Binet formula :-

Definition :-

Let 'A' be an $m \times n$ matrix such that

$$A = [C_1, C_2, \dots, C_n]_{m \times n} \text{ where } C_1, C_2, \dots, C_n \text{ are columns of 'A'}$$

and 'B' be an $n \times m$ matrix such that

$$B = \begin{bmatrix} R_1 \\ R_2 \\ \vdots \\ R_n \end{bmatrix}_{n \times m} \text{ where } R_1, R_2, R_3, \dots, R_n \text{ are rows of 'B'}$$

then

$$\det AB = \sum_{1 \leq j_1 < j_2 < \dots < j_m \leq n} \det [A_{C_{j_1}, C_{j_2}, \dots, C_{j_m}}] \cdot \det [B_{R_{j_1}, R_{j_2}, \dots, R_{j_m}}]$$

where: $m < n$

3:
Sol:

For Non-homogeneous System, write a given System of Eqn's in the form $AX=B$.
Later consider Augmented matrix $[A/B]$ reduce it into Echelon form. then find the $\rho[A]$ and $\rho[A/B]$.
 $\therefore$ If $\boxed{\rho[A] = \rho[A/B]}$ then the given non-homogeneous System of Eqn's are consistent.

4:
Sol:

Given System of Eqn's are

$$\left. \begin{array}{l} x + y + z = 6 \\ x + 2y + 2z = 14 \\ x + 4y + 7z = 30 \end{array} \right\} \rightarrow \text{①}$$

The matrix form of Eq ① is $AX=B$
Where

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 4 & 7 \end{bmatrix}; X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}; B = \begin{bmatrix} 6 \\ 14 \\ 30 \end{bmatrix}$$

Now consider Augmented Matrix $[A/B] \sim \begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & 2 & 2 & 14 \\ 1 & 4 & 7 & 30 \end{bmatrix}$

$$R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 1 & 8 \\ 0 & 3 & 6 & 24 \end{bmatrix} \quad R_3 \rightarrow R_3 - 3R_2$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 1 & 8 \\ 0 & 1 & 2 & 8 \end{bmatrix} \quad R_3 \rightarrow R_3 - R_2$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 1 & 8 \\ 0 & 0 & 1 & 0 \end{bmatrix} \rightarrow \text{②}$$

$\therefore$ Which is in Echelon form

No. of non zero rows = 3

$$\therefore \rho[A/B] = 3$$

Consider

$$A \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \therefore \text{which is in Echelon form}$$

No. of non zero rows = 3

$$\therefore \rho[A] = 3$$

$$\Rightarrow \rho[A] = 3 = \rho[A/B]$$

$$\boxed{\rho[A] = \rho[A/B]}$$

$\therefore$ The given system of Eqn's are consistent.

5. Rank of a Matrix :-

sol: If 'A' is a non-zero matrix we say that 'r' is Rank of A if :-

- i) Every $(r+1)^{\text{th}}$ order minor of A is '0' and
- ii) There exist, atleast one r^{th} order minor of A which is not zero.
- iii) Rank of A is denoted by ' $\rho[A]$ '.

6. Properties of Rank of a matrix :-

- sol:
- Rank of a Null matrix is '0'
 - If A is a non-singular matrix $[|A| \neq 0]$ then $\rho[A] \geq 1$
 - Rank of unit matrix of order 'n' is 'n'
 - If A is a matrix of order $m \times n$,
Rank of A is defined by $\therefore \rho[A] \leq \min(m, n)$
 - $|A| \neq 0$ then $\rho[A] = \text{order of matrix}$
 - $|A| = 0$ then $\rho[A] < \text{order of matrix}$
 - Rank of A and A^T is Same.

7. Normal form of a Matrix :-

- sol:
- By applying elementary row and column operations on the matrix A
 - It can be reduced to one of the following Normal form
 - The Normal form of a matrix are :-

$$N = I_r \text{ (or) } \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \text{ (or) } \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \text{ (or) } \begin{bmatrix} I_r \\ 0 \end{bmatrix}$$

8. Given matrix is :-

sol: let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & k & 7 \\ 3 & 6 & 10 \end{bmatrix}_{3 \times 3}$ given that Rank is '2'
order of matrix is '3'

wkt if $|A| = 0$ then $\rho[A]$ is less than the order of matrix

so,

$$|A| = 1(10k - 42) - 2(20 - 21) + 3(12 - 3k) = 0 \text{ (or)}$$

$$10k - 42 - 40 + 42 + 36 - 9k = 0$$

$$k - 40 + 36 = 0$$

$$k - 4 = 0$$

$$\boxed{k = 4}$$

$\therefore$ The value of k is '4'

$$\begin{aligned} & [A]_{3 \times 3} \\ & \downarrow \\ & |A| = 0 \\ & \downarrow \\ & \rho[A] < 3 \\ & \downarrow \\ & [A]_{2 \times 2} \\ & \downarrow \\ & |A| \neq 0 \\ & \downarrow \\ & \rho[A] = 2 \end{aligned}$$

9: Working procedure to reduce a given matrix into Echelon form are :-

By applying 'elementary row operations' on a matrix 'A' one can reduce it into Echelon form.

The matrix is said to be in Echelon form if :-

- i) The first non-zero entry in each non-zero row is equal to '1'.
 - ii) The zeroes in each row at the beginning is in increasing order.
 - iii) If any zero row occurs it should be last row of the matrix.
- Condition 'i' is optional

10: Given matrix is :-

sol

let $A = \begin{bmatrix} 8 & 1 & 3 & 6 \\ 0 & 3 & 2 & 2 \\ -8 & -1 & -3 & 4 \end{bmatrix}_{3 \times 4}$

The order of given matrix is 3×4
WKT

$$\begin{aligned} \rho(A) &\leq \min(m, n) \\ &\leq \min(3, 4) \\ &= 3 \end{aligned}$$

→ Now consider minor of $A = \begin{bmatrix} 8 & 1 & 3 \\ 0 & 3 & 2 \\ -8 & -1 & -3 \end{bmatrix}$

$$|A| = 8[-9+2] - 1[0+16] + 3[0+24]$$

$$|A| = 8(-7) - 1(16) + 3(24) = -56 - 16 + 72 = -72 + 72 = 0$$

$$\boxed{|A| = 0}$$

Now consider minor of $A = \begin{bmatrix} 8 & 1 \\ 0 & 3 \end{bmatrix}$

$$|A| = 24 - 0 = 24 \neq 0$$

$$\therefore |A| \neq 0$$

→ another minor of $A = \begin{bmatrix} 1 & 3 & 6 \\ 3 & 2 & 2 \\ -1 & -3 & 4 \end{bmatrix}_{3 \times 3}$

$$|A| = 1(8+6) - 3(12+2) + 6(-9+2)$$

$$|A| = 14 - 42 - 42 = -70$$

$$\therefore |A| \neq 0$$

So, order of given matrix is rank of the matrix

$$\therefore \boxed{\rho(A) = 3}$$

1: sol Rank of a matrix :-
The no. of non-zero rows in an Echelon form is the "Rank of a Matrix".

Given matrix is :-

$$\text{let } A = \begin{bmatrix} 3 & 1 & 4 & 6 \\ 2 & 1 & 2 & 4 \\ 4 & 2 & 5 & 8 \\ 1 & 1 & 2 & 2 \end{bmatrix} \quad R_1 \leftrightarrow R_4$$

$$\sim \begin{bmatrix} 1 & 1 & 2 & 2 \\ 2 & 1 & 2 & 4 \\ 4 & 2 & 5 & 8 \\ 3 & 1 & 4 & 6 \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 4R_1 \\ R_4 \rightarrow R_4 - 3R_1 \end{array}$$

$$\sim \begin{bmatrix} 1 & 1 & 2 & 2 \\ 0 & -1 & -2 & 0 \\ 0 & -2 & -3 & 0 \\ 0 & -2 & -2 & 0 \end{bmatrix} \quad \begin{array}{l} R_3 \rightarrow R_3 - 2R_2 \\ R_4 \rightarrow R_4 - 2R_2 \end{array}$$

$$\sim \begin{bmatrix} 1 & 1 & 2 & 2 \\ 0 & -1 & -2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & 0 \end{bmatrix} \quad R_4 \rightarrow R_4 - 2R_3$$

$$\sim \begin{bmatrix} 1 & 1 & 2 & 2 \\ 0 & -1 & -2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\therefore$ which is in Echelon form

No. of non-zero rows = 3

$$\therefore \boxed{\rho[A] = 3}$$

2: sol: Given matrix is :

$$\text{let } A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 4 & 1 & 2 & 1 \\ 3 & -1 & 1 & 2 \\ 1 & 2 & 0 & 1 \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow R_2 - 4R_1 \\ R_3 \rightarrow R_3 - 3R_1 \\ R_4 \rightarrow R_4 - R_1 \end{array}$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -7 & 6 & -11 \\ 0 & -7 & 4 & -7 \\ 0 & 0 & 1 & -2 \end{bmatrix} \quad R_3 \rightarrow R_3 - R_2$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -7 & 6 & -11 \\ 0 & 0 & -2 & 4 \\ 0 & 0 & 1 & -2 \end{bmatrix} \quad R_4 \rightarrow 2R_4 + R_3$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -7 & -6 & -11 \\ 0 & 0 & -2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} C_2 \rightarrow C_2 - 2C_1 \\ C_3 \rightarrow C_3 + C_1 \\ C_4 \rightarrow C_4 - 3C_1 \end{array}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -7 & -6 & -11 \\ 0 & 0 & -2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} C_3 \rightarrow 7C_3 - 6C_2 \\ C_4 \rightarrow 7C_4 - 11C_2 \end{array}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -7 & 0 & 0 \\ 0 & 0 & -14 & 28 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad R_3 \rightarrow \frac{R_3}{14}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -7 & 0 & 0 \\ 0 & 0 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad C_4 \rightarrow C_4 + 2C_3$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -7 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} C_2 \rightarrow \frac{C_2}{-7} \\ C_3 \rightarrow \frac{C_3}{-1} \end{array}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \sim \begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix}$$

$\therefore$ Which is in Normal form

Order of Identity matrix = 3

$$\therefore \boxed{\rho(A) = 3}$$

Q: Given matrix is :-
sol:

$$A = \begin{bmatrix} 1 & 2 & 3 & -2 \\ 2 & -2 & 1 & 3 \\ 3 & 0 & 4 & 1 \end{bmatrix}$$

$$\text{let } A = I_3 A I_4$$

$$\begin{bmatrix} 1 & 2 & 3 & -2 \\ 2 & -2 & 1 & 3 \\ 3 & 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - 3R_1$$

$$\begin{bmatrix} 1 & 2 & 3 & -2 \\ 0 & -6 & -5 & 7 \\ 0 & -6 & -5 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$C_2 \rightarrow C_2 - 2C_1, C_3 \rightarrow C_3 - 3C_1, C_4 \rightarrow C_4 + 2C_1$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -6 & -5 & 7 \\ 0 & -6 & -5 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -2 & -3 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -6 & -5 & 7 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -2 & -3 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$C_3 \rightarrow 6C_3 - 5C_2, C_4 \rightarrow 6C_4 + 7C_2$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -6 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -2 & -8 & -2 \\ 0 & 1 & -5 & 7 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

$$R_2 \rightarrow \frac{R_2}{-6}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1/3 & -1/6 & 0 \\ -1 & -1 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -2 & -8 & -2 \\ 0 & 1 & -5 & 7 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

$$\begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix} = PAQ$$

where

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 1/3 & -1/6 & 0 \\ -1 & -1 & 1 \end{bmatrix}; Q = \begin{bmatrix} 1 & -2 & -8 & -2 \\ 0 & 1 & -5 & 7 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

$$\Rightarrow |P| = \frac{1}{6} \neq 0$$

$$|Q| = 36 \neq 0$$

$\therefore P \& Q$ are Non-singular $\in$

$$PAQ = \begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix}$$

Order of Identity matrix = 2

$$\therefore \rho[A] = 2$$

4)
sol:

Given matrix is :-

$$\text{let } A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}$$

We write it as

$$A = I_n A$$

$$A = I_3 A$$

$$\begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$$R_2 \rightarrow R_2 - R_1 ; R_3 \rightarrow R_3 + 2R_1$$

$$\begin{bmatrix} 1 & 1 & 3 \\ 0 & 2 & -6 \\ 0 & -2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} A$$

$$R_1 \rightarrow 2R_1 - R_2 , R_3 \rightarrow R_3 + R_2$$

$$\begin{bmatrix} 2 & 0 & 12 \\ 0 & 2 & -6 \\ 0 & 0 & -4 \end{bmatrix} = \begin{bmatrix} 3 & -1 & 0 \\ -1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} A$$

$$R_1 \rightarrow \frac{R_1}{2} , R_2 \rightarrow \frac{R_2}{2} , R_3 \rightarrow \frac{R_3}{-4}$$

$$\begin{bmatrix} 1 & 0 & 6 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3/2 & -1/2 & 0 \\ -1/2 & 1/2 & 0 \\ -1/4 & -1/4 & -1/4 \end{bmatrix} A$$

$$R_1 \rightarrow R_1 - 6R_3 , R_2 \rightarrow R_2 + 3R_3$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 & 3/2 \\ -5/4 & -1/4 & -3/4 \\ -1/4 & -1/4 & -1/4 \end{bmatrix} A$$

This is of the form

$$[I_3] = BA$$

$$A^{-1} = B$$

$$\therefore A^{-1} = \begin{bmatrix} 3 & 1 & 3/2 \\ -5/4 & -1/4 & -3/4 \\ -1/4 & -1/4 & -1/4 \end{bmatrix}$$

5: The given system of equations are
 sol:
$$\begin{cases} x + y - 3z + 2w = 0 \\ 2x - y + 2z - 3w = 0 \\ 3x - 2y + z - 4w = 0 \\ -4x + y - 3z + w = 0 \end{cases} \rightarrow \textcircled{1}$$

The matrix form of Eq (1) is $AX=0$
 where

$$A = \begin{bmatrix} 1 & 1 & -3 & 2 \\ 2 & -1 & 2 & -3 \\ 3 & -2 & 1 & -4 \\ -4 & 1 & -3 & 1 \end{bmatrix} ; X = \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} ; 0 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

consider $[A] \sim \begin{bmatrix} 1 & 1 & -3 & 2 \\ 2 & -1 & 2 & -3 \\ 3 & -2 & 1 & -4 \\ -4 & 1 & -3 & 1 \end{bmatrix}$ $\begin{matrix} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1 \\ R_4 \rightarrow R_4 + 4R_1 \end{matrix}$

$$\sim \begin{bmatrix} 1 & 1 & -3 & 2 \\ 0 & -3 & 8 & -7 \\ 0 & -5 & 10 & -10 \\ 0 & 5 & -15 & 9 \end{bmatrix} \begin{matrix} R_3 \rightarrow 3R_3 - 5R_2 \\ R_4 \rightarrow 3R_4 + 5R_2 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 1 & -3 & 2 \\ 0 & -3 & 8 & -7 \\ 0 & 0 & -10 & 5 \\ 0 & 0 & -5 & -8 \end{bmatrix} \quad R_3 \rightarrow \frac{R_3}{5}$$

$$\sim \begin{bmatrix} 1 & 1 & -3 & 2 \\ 0 & -3 & 8 & -7 \\ 0 & 0 & -2 & 1 \\ 0 & 0 & 0 & -21 \end{bmatrix} \rightarrow \textcircled{A}$$

which is in Echelon form.

No. of non zero rows = 4

$$\therefore \rho(A) = 4 = r$$

We have $n = 4$ $[x, y, z, w]$ are unknowns $\therefore r = n = 4$

The given system has trivial solⁿ set.

$$\boxed{r=n}$$

From (A) By Backward substitution

$-21w = 0$ $\boxed{w=0}$	$-2z + w = 0$ $-2z = 0$ $\boxed{z=0}$	$-3y + 8z - 7w = 0$ $-3y + 0 - 0 = 0$ $-3y = 0$ $\boxed{y=0}$	$x + y - 3z + 2w = 0$ $x + 0 - 0 + 0 = 0$ $\boxed{x=0}$
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$$\therefore X = \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \text{ is the required trivial solⁿ set.}$$

G: A system of homogeneous linear equations in
 sol: n unknowns is a set of linear equations of
 the form $Ax=0$

consider the sym of m linear Equation in n unknowns
 $x_1, x_2, \dots, x_n$ as given below,

$$\left. \begin{aligned} 2x + y + 2z &= 0 \\ x + y + 3z &= 0 \\ 4x + 3y + kz &= 0 \end{aligned} \right\} \rightarrow \textcircled{1}$$

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= 0 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= 0 \\ \vdots & \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= 0 \end{aligned}$$

The matrix form of Eqn $\textcircled{1}$ is $Ax=0$
 where

$$A = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 1 & 3 \\ 4 & 3 & k \end{bmatrix}; \quad x = \begin{bmatrix} x \\ y \\ z \end{bmatrix}; \quad 0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The given sym posses a non trivial solution
 so, $\rho[A] < n$

Given $n=3$

i.e. $\rho[A] < 3$

i.e. $|A| = 0$

$$\begin{vmatrix} 2 & 1 & 2 \\ 1 & 1 & 3 \\ 4 & 3 & k \end{vmatrix} = 2(k-9) - 1(k-12) + 2(3-4) = 0$$

$$k-8=0$$

$$\boxed{k=8}$$

When $k=8$ then

$$A \sim \begin{bmatrix} 2 & 1 & 2 \\ 1 & 1 & 3 \\ 4 & 3 & 8 \end{bmatrix} \quad R_1 \leftrightarrow R_2$$

$$\sim \begin{bmatrix} 1 & 1 & 3 \\ 2 & 1 & 2 \\ 4 & 3 & 8 \end{bmatrix} \quad \begin{aligned} R_2 &\rightarrow R_2 - 2R_1 \\ R_3 &\rightarrow R_3 - 4R_1 \end{aligned}$$

$$\sim \begin{bmatrix} 1 & 1 & 3 \\ 0 & -1 & -4 \\ 0 & -1 & -4 \end{bmatrix} \quad R_3 \rightarrow R_3 - R_2$$

$$\sim \begin{bmatrix} 1 & 1 & 3 \\ 0 & -1 & -4 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \textcircled{A}$$

$\therefore$ which is in Echelon form

No. of non-zero rows = 2 = $r \therefore \rho[A] = 2 = r$

G.T $n=3$ [x, y, z unknowns]

$$r=2$$

$$r=2 < 3=n$$

$$\boxed{r < n}$$

The given system has non-trivial solution
i.e., infinite no. of solutions

from (A) Matrix form is

$$\begin{bmatrix} 1 & 1 & 3 \\ 0 & -1 & -4 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

choose,

$$n-r=3-2=1$$

consider '1' Arbitrary Constant

By Backward Substitution

i.e. $\boxed{z=k}$

$$-y - 4z = 0 \rightarrow (b)$$

sub z in (b)

$$-y - 4k = 0$$

$$\boxed{y = -4k}$$

$$x + y + 3z = 0 \rightarrow (c)$$

sub z, y in (c)

$$x + (-4k) + 3(k) = 0$$

$$x - 4k + 3k = 0$$

$$x - k = 0$$

$$\boxed{x = k}$$

$$\therefore X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} k \\ -4k \\ k \end{bmatrix} = k \begin{bmatrix} 1 \\ -4 \\ 1 \end{bmatrix} \text{ is the required Non-Trivial soln}$$

i.e. Infinite no. of solns
where 'k' is an Arbitrary Constant

7: The given system of Eqn's are:-

Sol:

$$\left. \begin{array}{l} x + y + 2z = 9 \\ x - 2y + 2z = 3 \\ 2x - y + z = 3 \\ 3x - y + z = 4 \end{array} \right\} \rightarrow (1)$$

The matrix form of Eqn (1) is $AX=B$
where

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & 2 \\ 2 & -1 & 1 \\ 3 & -1 & 1 \end{bmatrix} ; X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} ; B = \begin{bmatrix} 9 \\ 3 \\ 3 \\ 4 \end{bmatrix}$$

Consider Augmented matrix $[A/B] = \begin{bmatrix} 1 & 1 & 2 & 9 \\ 1 & -2 & 2 & 3 \\ 2 & -1 & 1 & 3 \\ 3 & -1 & 1 & 4 \end{bmatrix}$

$$R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - 2R_1, R_4 \rightarrow R_4 - 3R_1$$

$$\sim \begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & -3 & 0 & -6 \\ 0 & -3 & -3 & -15 \\ 0 & -4 & -5 & -23 \end{bmatrix} \quad R_2 \rightarrow \frac{R_2}{-3}, R_3 \rightarrow \frac{R_3}{-3}$$

$$\sim \begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & 0 & 2 \\ 0 & 1 & 1 & 5 \\ 0 & -4 & -5 & -23 \end{bmatrix} \quad R_3 \rightarrow R_3 - R_2, R_4 \rightarrow R_4 + 4R_2$$

$$\sim \begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & -5 & -15 \end{bmatrix} \quad R_4 \rightarrow \frac{R_4}{-5}$$

$$\sim \begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 1 & 3 \end{bmatrix} \quad R_4 \rightarrow R_4 - R_3$$

$$\sim \begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \textcircled{A}$$

$\therefore$ which is in Echelon form

No. of non-zero rows = 3

$$\therefore \rho[A/B] = 3$$

From $\textcircled{A}$

$$A \sim \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore$ which is in Echelon form

No. of non zero rows = 3

$$\therefore \rho[A] = 3$$

$\therefore \rho[A] = \rho[A/B] = 3 = r$ $\therefore$ which is consistent

Here $r = n = 3$

$$\boxed{r = n}$$

The given sys of Eqn's are having Unique solution

From (A) By Backward Substitution

$$z=3$$

$$y=2$$

$$x+y+2z=9 \rightarrow (b)$$

Sub x, y in (b)

$$x+2+2(3)=9$$

$$x+2+6=9$$

$$x+8=9$$

$$x=9-8=1$$

$$\therefore x=1$$

$$\therefore X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \text{ is the required soln set.}$$

8: The given sys of Eqn's are :-
sol:

$$\left. \begin{aligned} x+y+z &= 6 \\ x+2y+3z &= 10 \\ x+2y+\lambda z &= \mu \end{aligned} \right\} \rightarrow (1)$$

The matrix form of Eq (1) is

$$AX=B$$

where

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{bmatrix}; X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}; B = \begin{bmatrix} 6 \\ 10 \\ \mu \end{bmatrix}$$

Now consider Augmented matrix $[A/B] = \begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{bmatrix}$

$$R_2 \rightarrow R_2 - R_1; R_3 \rightarrow R_3 - R_1$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda-1 & \mu-6 \end{bmatrix} R_3 \rightarrow R_3 - R_2$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda-3 & \mu-10 \end{bmatrix} \rightarrow (A)$$

i) No solution $[s[A] \neq s[A/B]]$

If $\lambda=3, \mu \neq 10$ in (A)

$$[A/B] \sim \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 0 & \mu-10 \end{bmatrix}$$

which is in Echelon form

No. of non zero rows = 3

$$\therefore \cancel{S[A/B]} = 3$$

consider

$$A \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

which is in Echelon form

No. of non zero rows = 2

$$S[A] = 2$$

$$\therefore S[A] = 2 \neq 3 = S[A/B]$$

$$\therefore S[A] \neq S[A/B]$$

The given sys of Eqn's is inconsistent

It has No solution.

ii) Unique solution $[S[A] = S[A/B] = r = n]$
i.e. $n = 3$ [x, y, z are unknowns]

If $\lambda \neq 3$, $\mu \neq 10$ in (A)

$$[A/B] \sim \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda-3 & \mu-10 \end{bmatrix}$$

Which is in Echelon form

No. of non zero rows = 3 $\therefore S[A/B] = 3$

Consider

$$A \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & \lambda-3 \end{bmatrix}$$

which is in Echelon form

No. of non zero rows = 3

$$\therefore S[A] = 3$$

$$\therefore S[A] = S[A/B] = 3 = r = n$$

$$r = n$$

The given system of Eqn's is Consistent

It has an Unique solution.

iii) An Infinite no. of solutions $[S[A] = S[A/B] = r < n]$

i.e. $n = 3$ [x, y, z are unknowns] ; $r = ?$

If $\lambda = 3$, $\mu = 10$ in (A)

$$[A/B] \sim \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 9 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\therefore$ which is in Echelon form
No. of non zero rows = 2 $\therefore \rho[A/B] = 2$

consider

$$A \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \therefore \text{which is in Echelon form}$$

No. of non zero rows = 2

$$\therefore \rho[A] = 2$$

$\therefore \rho[A] = \rho[A/B] = r = 2$ which is consistent

where $r = 2, n = 3$

$$r = 2 < 3 = n$$

$$\boxed{r < n}$$

$\therefore$ The given system of Eqn's having infinite no. of soln's

q:
sol

The given system of Eqn's are :-

$$\left. \begin{aligned} 3x + y - z &= 3 \\ 2x - 8y + z &= -5 \\ x - 2y + 9z &= 8 \end{aligned} \right\} \rightarrow \textcircled{1}$$

The matrix form of Eq① is

$$AX = B$$

where

$$A = \begin{bmatrix} 3 & 1 & -1 \\ 2 & -8 & 1 \\ 1 & -2 & 9 \end{bmatrix} ; X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} ; B = \begin{bmatrix} 3 \\ -5 \\ 8 \end{bmatrix}$$

consider Augmented matrix $[A/B] \sim \begin{bmatrix} 3 & 1 & -1 & 3 \\ 2 & -8 & 1 & -5 \\ 1 & -2 & 9 & 8 \end{bmatrix}$

$$R_1 \leftrightarrow R_3$$

$$\sim \begin{bmatrix} 1 & -2 & 9 & 8 \\ 2 & -8 & 1 & -5 \\ 3 & 1 & -1 & 3 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1 \end{array}$$

$$\sim \begin{bmatrix} 1 & -2 & 9 & 8 \\ 0 & -4 & -17 & -21 \\ 0 & 7 & -28 & -21 \end{bmatrix} \begin{array}{l} \\ R_3 \rightarrow \frac{R_3}{7} \end{array}$$

$$\sim \begin{bmatrix} 1 & -2 & 9 & 8 \\ 0 & -4 & -17 & -21 \\ 0 & 1 & -4 & -3 \end{bmatrix} \quad R_3 \rightarrow 4R_3 + R_2$$

$$\sim \begin{bmatrix} 1 & -2 & 9 & 8 \\ 0 & -4 & -17 & -21 \\ 0 & 0 & -33 & -33 \end{bmatrix} \rightarrow \textcircled{A}$$

$\therefore$ which is in Echelon form

By Backward Substitution

The matrix form is

$$\begin{bmatrix} 1 & -2 & 9 \\ 0 & -4 & -17 \\ 0 & 0 & -33 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ -21 \\ -33 \end{bmatrix}$$

we have,

$$-33z = -33$$

$$\boxed{z = 1}$$

$$-4y - 17z = -21 \rightarrow \textcircled{b}$$

sub z in $\textcircled{b}$

$$-4y - 17(1) = -21$$

$$-4y = -21 + 17$$

$$-4y = -4$$

$$\boxed{y = 1}$$

$$x - 2y + 9z = 8 \rightarrow \textcircled{c}$$

sub z, y in $\textcircled{c}$

$$x - 2(1) + 9(1) = 8$$

$$x = 8 - 9 + 2$$

$$\boxed{x = 1}$$

$$\therefore X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \text{ is the required soln set.}$$

10:
Sol:

The given sys of Eqn's are:-

$$10x - 5y - 2z = 3 \Rightarrow |10| \quad |5| + |2| \Rightarrow 10 > 5 + 2$$

$$4x - 10y + 3z = -3 \Rightarrow |10| \quad |4| + |3| \Rightarrow 10 > 4 + 3$$

$$x + 6y + 10z = -3 \Rightarrow |10| \quad |1| + |6| \Rightarrow 10 > 1 + 6$$

The given sys of Eqn's are Diagonally Dominant
we write as:-

$$10x = 3 + 5y + 2z \Rightarrow x = \frac{1}{10} [3 + 5y + 2z] \rightarrow \textcircled{1}$$

$$-10y = -3 - 4x - 3z \Rightarrow y = \frac{1}{10} [3 + 4x + 3z] \rightarrow \textcircled{2}$$

$$10z = -3 - x - 6y \Rightarrow z = \frac{1}{10} [-3 - x - 6y] \rightarrow \textcircled{3}$$

Iteration - 1

let $y=0, z=0$ in ①

$$x^{(1)} = \frac{1}{10} [3 + 5(0) + 2(0)] = 0.3$$

put $x = x^{(1)} = 0.3, z=0$ in ②

$$y^{(1)} = \frac{1}{10} [3 + 4(0.3) + 3(0)] = 0.42$$

put $x = x^{(1)} = 0.3, y = y^{(1)} = 0.42$ in ③

$$z^{(1)} = \frac{1}{10} [-3 - (0.3) - 6(0.42)] = -0.582$$

Iteration - 2

put $y = y^{(1)} = 0.42, z = z^{(1)} = -0.582$ in ①

$$x^{(2)} = \frac{1}{10} [3 + 5(0.42) + 2(-0.582)] = 0.394$$

put $x = x^{(2)} = 0.394, z = z^{(1)} = -0.582$ in ②

$$y^{(2)} = \frac{1}{10} [3 + 4(0.394) + 3(-0.582)] = 0.283$$

put $x = x^{(2)} = 0.394, y = y^{(2)} = 0.283$ in ③

$$z^{(2)} = \frac{1}{10} [-3 - (0.394) - 6(0.283)] = -0.509$$

Iteration - 3

put $y = y^{(2)} = 0.283, z = z^{(2)} = -0.509$ in ①

$$x^{(3)} = \frac{1}{10} [3 + 5(0.283) + 2(-0.509)] = 0.331$$

put $x = x^{(3)} = 0.331, z = z^{(2)} = -0.509$ in ②

$$y^{(3)} = \frac{1}{10} [3 + 4(0.331) + 3(-0.509)] = 0.271$$

put $x = x^{(3)} = 0.331, y = y^{(3)} = 0.271$ in ③

$$z^{(3)} = \frac{1}{10} [-3 - (0.331) - 6(0.271)] = -0.496$$

Iteration - 4

put $y = y^{(3)} = 0.271, z = z^{(3)} = -0.496$ in ①

$$x^{(4)} = \frac{1}{10} [3 + 5(0.271) + 2(-0.496)] = 0.336$$

put $x = x^{(4)} = 0.336, z = z^{(3)} = -0.496$ in ②

$$y^{(4)} = \frac{1}{10} [3 + 4(0.336) + 3(-0.496)] = 0.286$$

put $x = x^{(4)} = 0.336, y = y^{(4)} = 0.286$ in ③

$$z^{(4)} = \frac{1}{10} [-3 - (0.336) - 6(0.286)] = -0.505$$

Iteration - 5

put $y = y^{(4)} = 0.286$, $z = z^{(4)} = -0.505$ in (1)

$$x^{(5)} = \frac{1}{10} [3 + 5(0.286) + 2(-0.505)] = 0.342$$

put $x = x^{(5)} = 0.342$, $z = z^{(4)} = -0.505$ in (2)

$$y^{(5)} = \frac{1}{10} [3 + 4(0.342) + 3(-0.505)] = 0.285$$

put $x = x^{(5)} = 0.342$, $y = y^{(5)} = 0.285$ in (3)

$$z^{(5)} = \frac{1}{10} [-3 - (0.342) - 6(0.285)] = -0.505$$

Iteration - 6

put $y = y^{(5)} = 0.285$, $z = z^{(5)} = -0.505$ in (1)

$$x^{(6)} = \frac{1}{10} [3 + 5(0.285) + 2(-0.505)] = 0.342$$

put $x = x^{(5)} = 0.342$, $z = z^{(5)} = -0.505$ in (2)

$$y^{(6)} = \frac{1}{10} [3 + 4(0.342) + 3(-0.505)] = 0.285$$

put $x = x^{(5)} = 0.342$, $y = y^{(5)} = 0.285$ in (3)

$$z^{(6)} = \frac{1}{10} [-3 - (0.342) - 6(0.285)] = -0.505$$

$\therefore$ Since the values of 5th & 6th Iterations are similar.

$$\boxed{x = 0.342}, \boxed{y = 0.285}, \boxed{z = -0.505}$$

We tabulate the result as below :-

Variables	Iteration 1	I-2	I-3	I-4	I-5	I-6
x	0.3	0.394	0.331	0.336	0.342	0.342
y	0.42	0.283	0.271	0.286	0.285	0.285
z	-0.582	-0.509	-0.496	-0.505	-0.505	-0.505

11:
Sol.

The given system of Eqn's are:-

$$8x - 3y + 2z = 20 \Rightarrow |8| > | -3| + |2| \Rightarrow 8 > 5$$

$$4x + 11y - z = 33 \Rightarrow |11| > |4| + |-1| \Rightarrow 11 > 5$$

$$6x + 3y + 12z = 35 \Rightarrow |12| > |6| + |3| \Rightarrow 12 > 9$$

The given system of Eqn's are Diagonally Dominant
We write it as

$$\left. \begin{aligned} 8x &= 20 + 3y - 2z \Rightarrow x = \frac{1}{8} [20 + 3y - 2z] \\ 11y &= 33 - 4x + z \Rightarrow y = \frac{1}{11} [33 - 4x + z] \\ 12z &= 35 - 6x - 3y \Rightarrow z = \frac{1}{12} [35 - 6x - 3y] \end{aligned} \right\} \rightarrow \textcircled{1}$$

Iteration - 1

put $x=0, y=0, z=0$ in $\textcircled{1}$

$$x^{(1)} = \frac{1}{8} [20 + 3(0) - 2(0)] = 2.5$$

$$y^{(1)} = \frac{1}{11} [33 - 4(0) + 0] = 3.0$$

$$z^{(1)} = \frac{1}{12} [35 - 6(0) - 3(0)] = 2.917$$

Iteration - 2

put $x=x^{(1)}=2.5, y=y^{(1)}=3.0, z=z^{(1)}=2.917$ in $\textcircled{1}$

$$x^{(2)} = \frac{1}{8} [20 + 3(3) - 2(2.917)] = 2.896 \approx 2.90$$

$$y^{(2)} = \frac{1}{11} [33 - 4(2.5) + 2.917] = 2.3561$$

$$z^{(2)} = \frac{1}{12} [35 - 6(2.5) - 3(3)] = 0.917 \approx 0.92$$

Iteration - 3 :-

put $x=x^{(2)}=2.90, y=y^{(2)}=2.3561, z=z^{(2)}=0.92$ in $\textcircled{1}$

$$x^{(3)} = \frac{1}{8} [20 + 3(2.3561) - 2(0.92)] = 3.154 \approx 3$$

$$y^{(3)} = \frac{1}{11} [33 - 4(2.90) + 0.92] = 2.0291 \approx 2$$

$$z^{(3)} = \frac{1}{12} [35 - 6(2.90) - 3(2.3561)] = 0.8776 \approx 0.9 \approx 1$$

Iteration - 4

put $x = x^{(3)} = 3.154$, $y = y^{(3)} = 2.0291$, $z = z^{(2)} = 0.9$ in ①

$$x^{(4)} = \frac{1}{8} [20 + 3(2.0291) - 2(0.9)] = 3.04 \approx 3$$

$$y^{(4)} = \frac{11}{11} [33 - 4(3.154) + 0.9] = 1.94 \approx 2$$

$$z^{(4)} = \frac{1}{12} [35 - 6(3.154) - 3(2.0291)] = 0.832 \approx 1$$

∴ Since the values of 3rd & 4th Iterations are similar

$$\boxed{x=3}, \quad \boxed{y=2}, \quad \boxed{z=1}$$

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